Problem 1 (2 pts)

Consider the following non-linear equation:

$$f(x) = x^2 + x - 1 = 0 \quad \text{on } [0, 1]$$

(a) Consider the bisection algorithm starting with the interval $[0, 1]$. Find the minimum number of iterations N required to approximate the solution with an absolute error of less than 10^{-5}.

(b) Then consider the stopping criterion as when the interval length is smaller than 10^{-3}. List the sequences a_n, b_n, $x_n = (a_n + b_n)/2$, and $f(x_n)$ in each step.

(a) We have shown in class that the error, $|x_n - x^*|$, is bounded such that, $|x_n - x^*| \leq \frac{1}{2^{n+1}}(b-a)$ on the interval $[a, b]$. Thus using our interval of $[0, 1]$ we know that after n steps of the bisection algorithm the error is bounded by $\frac{1}{2^{n+1}}$. Set this error equal to the desired resolution of 10^{-5} and we have, $2^{n+1} = 10^5$ thus $n = \frac{5}{\log_{10} 2} - 1 \approx 15.609$. Thus $\lceil n \rceil = 16$ gives us the minimum amount of steps required.

(b) We see from part a that the minimum number of steps will be $\lceil \frac{3}{\log_{10} 2} - 1 \rceil = 9$. To avoid any arithmetic errors, I wrote an Rscript (see code appendix) to run the bisection algorithm for me and save the sequences. The sequences are all rounded to 4 decimal places as our goal is to have an error no more than 10^{-3}.

n	a_n	b_n	x_n	$f(x_n)$
0	0.0000	1.0000	0.5000	-0.2500
1	0.5000	1.0000	0.7500	0.3125
2	0.5000	0.7500	0.6250	0.0156
3	0.5000	0.6250	0.5625	-0.1211
4	0.5625	0.6250	0.5938	-0.0537
5	0.5938	0.6250	0.6094	-0.0193
6	0.6094	0.6250	0.6172	-0.0019
7	0.6172	0.6250	0.6211	0.0069
8	0.6172	0.6211	0.6191	0.0025
9	0.6172	0.6191	0.6182	0.0003

Table 1: Sequences

Problem 2 (7 pts)

Consider the following non-linear equation:

$$f(x) = x^2 + x - 1 = 0$$

(a) Write down the updating equation of f of the sequence $\{p_n\}_{n=1}^{\infty}$ by Newton's method.

(b) Let $x_0 = 1$ and assume that the sequence $\{p_n\}_{n=1}^{\infty}$ converges (this assumption only holds in this sub-question). Show <u>analytically</u> that $\{p_n\}_{n=1}^{\infty}$ converges to $\frac{-1 \pm \sqrt{5}}{2}$ by direction computation. (Notice that $p_n \geq 0$, so you might have to reject an answer that you calculated.)

(c) **Computational problem.** Solve the equation by Newton's method with $x_0 = 1$. Let your code print the x_n's generated by Newton's method, until a stopping criteria is reached. Take a screenshot of the results printed by your code, submit BOTH your code and the screenshot of the computation results.

(d) Show that there exists $\delta > 0$ such that the sequence converges to $\frac{-1+\sqrt{5}}{2}$ if $p_0 \in \left[\frac{-1+\sqrt{5}}{2} - \delta, \frac{-1+\sqrt{5}}{2} + \delta\right]$.

(e) Show that we can choose δ such that, with such $p_0 \in \left[\frac{-1+\sqrt{5}}{2} - \delta, \frac{-1+\sqrt{5}}{2} + \delta\right]$, the sequence converges to $\frac{-1+\sqrt{5}}{2}$ at least quadratically.

(a) The updating equation by definition is,

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})} \quad \text{for } n \geq 1,$$
$$p_n = p_{n-1} - \frac{(p_{n-1})^2 + p_{n-1} - 1}{2p_{n-1} + 1} \quad \text{for } n \geq 1.$$

(b) Suppose that when $p_0 = 1$, the $p_n \to p^*$. Thus p_{n-1} also converges to p^*. Then from the updating equation of part (a) we get the following equation,

$$\begin{aligned} p^* &= p^* - \frac{(p^*)^2 + p^* - 1}{2p^* + 1} \\ \implies 0 &= \frac{(p^*)^2 + p^* - 1}{2p^* + 1} \\ \implies 0 &= (p^*)^2 + p^* - 1 \\ \implies p^* &= \frac{-1 \pm \sqrt{5}}{2}. \end{aligned}$$

Since $p_n \geq 0$ then p^* (by proximity of our starting point) must be the positive root, $\frac{-1+\sqrt{5}}{2}$.

(c) Refer to screenshot for sequence. Code responsible for the sequence can also be found in the appendix.

```
+   return(x_seq)
+ }
> ## Expression for Problem 2
> f <- expression(x^2+x-1)
> ## Calling NM function
> Newt(f, 1, 1e-6)
[1] 1.0000000 0.6666667 0.6190476 0.6180344 0.6180340
>
```

Figure 1: Screenshot of Sequence Generated by the Newton's Method Algorithm

(d) Let $\phi(x)$ be the iteration mapping such that $x_{n+1} = \phi(x_n)$ where $\phi(x) := x - \frac{f(x)}{f'(x)}$. Let $\alpha = \frac{-1+\sqrt{5}}{2}$,

then $2\alpha + 1 = \sqrt{5}$ and $\alpha^2 = \frac{6-2\sqrt{5}}{4}$. We first verify that $\phi(\alpha) = \alpha$.

$$\begin{aligned}
\phi(x) &= x - \frac{x^2 + x - 1}{2x + 1} \\
\implies \phi(\alpha) &= \alpha - \frac{\alpha^2 + \alpha - 1}{2\alpha + 1} \\
&= \frac{2\alpha^2 + \alpha - \alpha^2 - \alpha + 1}{2\alpha + 1} = \frac{\alpha^2 + 1}{2\alpha + 1} \\
&= \frac{1}{\sqrt{5}}(\alpha^2 + 1) = \frac{1}{\sqrt{5}}\left(\frac{6 - 2\sqrt{5}}{4} + \frac{4}{4}\right) \\
&= \frac{10 - 2\sqrt{5}}{4\sqrt{5}} = \frac{5 - \sqrt{5}}{2\sqrt{5}} = \frac{5/\sqrt{5} - 1}{2} = \frac{\sqrt{5} - 1}{2} = \alpha.
\end{aligned}$$

So indeed ϕ has α as a fixed point. Next consider $\phi'(x)$ at α.

$$\begin{aligned}
\phi'(x) &= 1 - \frac{2x^2 + 2x + 3}{(2x + 1)^2} \\
\implies \phi'(\alpha) &= 1 - \frac{2\alpha^2 + 2\alpha + 3}{(2\alpha + 1)^2} \\
&= 1 - \frac{2\alpha^2 + 2 + \sqrt{5}}{(\sqrt{5})^2} \\
&= 1 - \frac{\left(\frac{6-2\sqrt{5}}{2} + \frac{4}{2} + \frac{2\sqrt{5}}{2}\right)}{5} = 1 - \frac{10/2}{5} = 0.
\end{aligned}$$

Note that $\phi'(x)$ is locally continuous about α and $|\phi'(\alpha)| < 1$ therefore there exists a $\delta > 0$ such that $|\phi'(x)| \leq q < 1$ for all $x \in (\alpha - \delta, \alpha + \delta)$. Now we invoke the result from problem (4a) and note that so long as $x_0 \in \left[\frac{-1+\sqrt{5}}{2} - \delta, \frac{-1+\sqrt{5}}{2} + \delta\right]$, $x_n \to \alpha$ at least linearly.

(e) Using the definition from part (d) we note that $\phi'(\alpha) = 0$ and to use the Taylor series expansion about α we need to compute $\phi''(x)$. Computing we see that,

$$\begin{aligned}
\phi''(x) &= \frac{10}{(2x + 1)^3} \\
\implies \phi''(\alpha) &\approx 0.89443... \neq 0.
\end{aligned}$$

Now expand using the Taylor series expansion about α.

$$
\begin{aligned}
\phi(x_n) &= \phi(\alpha) + \cancelto{0}{\phi'(\alpha)}(x_n - \alpha) + \frac{\phi''(\alpha)}{2!}(x_n - \alpha)^2 + o((x_n - \alpha)^3) \\
\implies \phi(x_n) - \phi(\alpha) &= C(x_n - \alpha)^2 + o((x_n - \alpha)^3) \\
\implies \frac{|x_{n+1} - \alpha|}{|x_n - \alpha|^2} &= C + o(|x_n - \alpha|)
\end{aligned}
$$

where we let $C := \frac{\phi''(\alpha)}{2!}$. Now we take the limit as $n \to \infty$ of both sides and note that as x_n approaches α, $x_n - \alpha$ approaches 0, and thus $|x_n - \alpha|$ approaches zero. Therefore in the limit we have,

$$
\lim_{n\to\infty} \frac{|x_{n+1} - \alpha|}{|x_n - \alpha|^2} = C > 0.
$$

Therefore x_n converges to α at least quadratically.

Problem 3 (3 pts)

Consider the following function:

$$
f(x) = 1 - \cos(x) \quad \text{on } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]
$$

(a) Write down the sequence of Netwon's Method $\{x_n\}_{n=0}^{\infty}$,

(b) If $x_0 \in \{x \in \mathbb{R} \mid 2x \sin x + \cos x - 1 = 0\}$ and $x_0 \neq 0$, show that the sequence, with such an initial value, does not converge.

(a) First we compute $f'(x) = \sin x$. Then the sequence of Newton's Method is,

$$
x_n = x_{n-1} - \frac{1 - \cos(x_{n-1})}{\sin(x_{n-1})}.
$$

(b) Suppose $x_0 \in \{x \in \mathbb{R} \mid 2x \sin x + \cos x - 1 = 0\} \implies 1 - \cos(x_0) = 2x_0 \sin(x_0)$. Thus computing x_1 we get,

$$
\begin{aligned}
x_1 &= x_0 - \frac{1 - \cos x_0}{\sin x_0} = x_0 - \frac{2x_0 \sin x_0}{\sin x_0} = -x_0, \\
\implies x_2 &= -x_0 - \frac{1 - \cos(-x_0)}{\sin(-x_0)} = -x_0 + \frac{2x_0 \sin(x_0)}{\sin(x_0)} = x_0 \\
\implies x_3 &= -x_0 \\
\implies x_4 &= x_0 \\
&\vdots
\end{aligned}
$$

We see that in general picking such an initial value will result in the sequence $x_n = (-1)^n x_0$ for

$n \geq 1$. We note that this alternating sequence clearly does not converge to any value other than when $x_0 = 0$, however by assumption $x_0 \neq 0$.

Problem 4 (4 pts)

Prove the following theorems:

(a) **Theorem:** Let $\phi \in C^1([a,b])$. Assume that there exists $x^* \in (a,b)$ such that $\phi(x^*) = x^*$, and that $|\phi'(x^*)| < 1$. Then there exists $\varepsilon > 0$ such that if $x_0 \in (x^* - \varepsilon, x^* + \varepsilon)$, then the sequence

$$x_{n+1} = \phi(x_n)$$

converges to x^* at least linearly.

(b) **Theorem:** Let $\phi \in C^k([a,b])$. Assume that there exists $x^* \in (a,b)$ such that $\phi(x^*) = x^*$, and there exists k such that $\phi^{(i)}(x^*) = 0$ for all $i < k$. Then there exists $\varepsilon > 0$ such that if $x_0 \in (x^* - \varepsilon, x^* + \varepsilon)$, then the sequence

$$x_{n+1} = \phi(x_n)$$

converges to x^* at least with order k.

(a) Note that ϕ is C^1 on $[a,b]$ and for some $x^* \in (a,b)$ such that $|\phi'(x^*)| < 1$ implies that there exists $\varepsilon > 0$ such that $|\phi'(x)| \leq q < 1$ for some $q \in (0,1)$ and for all $x \in [x^* - \varepsilon, x^* + \varepsilon]$. Note too that ϕ is a contraction mapping on the interval $[x^* - \varepsilon, x^* + \varepsilon]$. Let $x_0 \in [x^* - \varepsilon, x^* + \varepsilon]$ we will show that $\phi(x_0)$ is contained within the same interval. Consider,

$$\begin{aligned}
&|x_1 - x^*| = |\phi(x_0) - \phi(x^*)| \quad \text{by the mean value theorem } \exists \eta_0 \in (x_0, x^*) \text{ such that,} \\
&|x_1 - x^*| = \phi'(\eta_0)|x_0 - x^*| \\
\implies &|x_1 - x^*| \leq q|x_0 - x^*|.
\end{aligned}$$

Therefore x_1 is closer to x^* than x_0 is and thus $x_1 \in [x^* - \varepsilon, x^* + \varepsilon]$. Thus $\mathrm{Im}\,(\phi) \subseteq [x^* - \varepsilon, x^* + \varepsilon]$. Thus we can be certain that $x_n \in [x^* - \varepsilon, x^* + \varepsilon]$ for all $n \geq 1$ so long as $x_0 \in [x^* - \varepsilon, x^* + \varepsilon]$.

Now consider the following ratio,

$$\begin{aligned}
&\frac{|x_{n+1} - x^*|}{|x_n - x^*|} = \frac{\phi'(\eta_n)|x_n - x^*|}{|x_n - x^*|} \quad \text{(by Mean Value Theorem and for some } \eta_n \in (x_n, x^*)) \\
\implies &\frac{|x_{n+1} - x^*|}{|x_n - x^*|} \leq q \\
\implies &\lim_{n\to\infty} \frac{|x_{n+1} - x^*|}{|x_n - x^*|} \leq \lim_{n\to\infty} q \\
\implies &\lim_{n\to\infty} \frac{|x_{n+1} - x^*|}{|x_n - x^*|} \leq q \in [0,1).
\end{aligned}$$

Thus by definition x_n converges to x^* at least linearly.

(b) Consider the Taylor series approximation of $\phi(x_n)$ about x^* using $k+1$ terms.

$$\phi(x_n) \approx \phi(x^*) + \phi'(x^*)(x_n - x^*) + \cdots + \frac{\phi^{(k)}(x^*)}{k!}(x_n - x^*)^k$$
$$\phi(x_n) = \phi(x^*) + \phi'(x^*)(x_n - x^*) + \cdots + \frac{\phi^{(k)}(\eta_n)}{k!}(x_n - x^*)^k,$$

by the mean value theorem for some $\eta_n \in (x_n, x^*)$. By assumption all terms with i^{th} derivatives for $i < k$ are zero. Thus we get the simplified expression

$$\begin{aligned}
\phi(x_n) &= \phi(x^*) + \frac{\phi^{(k)}(\eta_n)}{k!}(x_n - x^*)^k \\
\implies \phi(x_n) - \phi(x^*) &= \frac{\phi^{(k)}(\eta_n)}{k!}(x_n - x^*)^k \\
\implies |\phi(x_n) - \phi(x^*)| &= \left|\frac{\phi^{(k)}(\eta_n)}{k!}(x_n - x^*)^k\right| \\
\implies \frac{|\phi(x_n) - \phi(x^*)|}{|x_n - x^*|^k} &= \left|\frac{\phi^{(k)}(\eta_n)}{k!}\right| \\
\implies \frac{|x_{n+1} - x^*|}{|x_n - x^*|^k} &= \frac{|\phi^{(k)}(\eta_n)|}{k!} \\
\implies \lim_{n\to\infty} \frac{|x_{n+1} - x^*|}{|x_n - x^*|^k} &= \lim_{n\to\infty} \frac{|\phi^{(k)}(\eta_n)|}{k!} = \frac{1}{k!}|\phi^{(k)}(\lim_{n\to\infty} \eta_n)| \quad (\text{since } \phi \in C^k([a,b])) \\
\implies \lim_{n\to\infty} \frac{|x_{n+1} - x^*|}{|x_n - x^*|^k} &= \frac{\phi^{(k)}(x^*)}{k!} < \infty.
\end{aligned}$$

Note that $\eta_n \in (x_n, x^*)$ and $x_n \to x^*$ thus by the squeeze theorem $\eta_n \to x^*$. Therefore this shows that x_n converges to x^* with order k.

Problem 5 (4 pts)

Prove the Fixed Point Theorem and its corollary:

(a) **Fixed Point Theorem:** Let $g(x) \in C([a,b])$ such that $g(x) \in [a,b]$ for any $x \in [a,b]$ and $g \in C^1([a,b])$. There exists $k \in (0,1)$ such that $|g'(x)| \leq k < 1$ for any $x \in (a,b)$. Then for any $x_0 \in [a,b]$, the sequence $\{x_n\}$ defined by

$$x_n = g(x_{n-1}), \quad n \geq 1$$

converges to the unique fixed point of g in $[a,b]$.

(b) **Corollary:** The error bound in the fixed-point iteration follows

$$|x_n - x^*| \leq k^n \max\{|x_0 - a|, |x_0 - b|\},$$

and

$$|x_n - x^*| \leq \frac{k^n}{1-k}|x_1 - x_0|, \quad \forall n \geq 1.$$

(a) Note that g is a contraction mapping on $[a, b]$ and by the extra condition that it is C^1 on $[a, b]$ and has a derivative strictly less then 1 on the interval then by the theorem covered in class on contraction mappings we are guaranteed a unique fixed point $x^* \in [a, b]$ such that $g(x^*) = x^*$. Let $e_n := x_n - x^*$, then $e_{n+1} = x_{n+1} - x^* = g(x_n) - x^* = g(x_n) - g(x^*)$. Thus by the mean value theorem there exists an $\eta_n \in (x_n, x^*)$ such that $e_{n+1} = g'(\eta_n)(x_n - x^*)$, or more simply $e_{n+1} = g'(\eta_n)e_n$. We continue this recurrence relation and see that,

$$\begin{aligned} e_{n+1} &= g'(\eta_n)e_n = g'(\eta_n)g'(\eta_{n-1})e_{n-1} = \cdots = \prod_{i=0}^{n} g'(\eta_i)e_0 \\ \implies |e_{n+1}| &\leq k^{n+1}|e_0|. \end{aligned}$$

This last line follows from the assumption that $|g'(x)| \leq k < 1$ for all $x \in (a, b)$. Since $k < 1$ there exists $p > 1$ such that $\frac{1}{p} := k$. Thus $|e_{n+1}| = \left(\frac{1}{p}\right)^{n+1}|e_0|$ and it follows that,

$$\begin{aligned} &\lim_{n\to\infty} |e_{n+1}| = \lim_{n\to\infty} \frac{1}{p^{n+1}}|e_0| = 0 \\ \implies &|e_n| \to 0 \iff e_n \to 0 \iff x_n \to x^* \iff g(x_n) \to x^*. \end{aligned}$$

(b) From part (a) we found the error bound $|x_n - x^*| \leq k^n|x_0 - x^*|$. Note that $[a, b] = [a, x^*] \cup (x^*, b]$ and $x_0 \in [a, b]$. If $x_0 \in [a, x^*]$, then $|x_0 - x^*| \leq |x_0 - b|$ otherwise if $x_0 \in (x^*, b]$, then $|x_0 - x^*| \leq |x_0 - a|$. Thus $|x_0 - x^*| \leq \max\{|x_0 - a|, |x_0 - b|\}$. Therefore,

$$|x_n - x^*| \leq k^n|x_0 - x^*| \leq k^n \max\{|x_0 - a|, |x_0 - b|\}.$$

Showing the second inequality was a little trickier than the first. First consider $|x_{n+1} - x_n|$, note that

$$\begin{aligned} &|x_{n+1} - x_n| = |g(x_n) - g(x_{n-1})| \leq k|x_n - x_{n-1}| \quad \text{(by what was shown in part a)} \\ &|x_{n+1} - x_n| \leq k^n|x_1 - x_0| \quad \text{(by iteration)} \end{aligned}$$

Let $m \in \mathbb{N}$ and consider

$$
\begin{aligned}
|x_{n+m} - x_n| &= |(x_{n+m} - x_{n+m-1}) + (x_{n+m-1} - x_{n+m-2}) + \cdots + (x_{n+1} - x_n)| \\
&\leq \underbrace{|x_{n+m} - x_{n+m-1}| + \cdots + |x_{n+1} - x_n|}_{m \text{ terms}} \qquad \text{(by the triangle inequality)} \\
|x_{n+m} - x_n| &\leq k^{n+m}|x_1 - x_0| + k^{n+m-1}|x_1 - x_0| + \cdots + k^n|x_1 - x_0| \\
\implies |x_{n+m} - x_n| &\leq \left(\sum_{i=0}^{m} k^i\right) k^n|x_1 - x_0| \\
\implies \lim_{m\to\infty} |x_{n+m} - x_n| &\leq \lim_{m\to\infty} \left(\sum_{i=0}^{m} k^i\right) k^n|x_1 - x_0|.
\end{aligned}
$$

Note that the left hand side of the inequality converges to $|x^* - x_n|$ and the right hand side is a geometric sum, and with $k < 1$ it converges to $\frac{k^n}{1-k}|x_1 - x_0|$. Therefore we have,

$$
|x_n - x^*| \leq \frac{k^n}{1-k}|x_1 - x_0|.
$$

Code Appendix

Both algorithms are written in R.

Bisection Algorithm

```r
## Bisection Algorithm ##

G <- function(x){
  return(x^2 +x -1)
}

Bis_Alg <- function(H, a, b, max_steps){
  #H is the function and [a,b] is the interval.
  if(a < b){
    if(H(a)*H(b)<0){
      #init counter and interval
      i <- 0
      A <- a
      B <- b

      a_seq <- c()
      b_seq <- c()
      x_seq <- c()
      f_seq <- c()
      while (i <= max_steps){
        x <- (A + B)/2
        f <- H(x)

        # Saving the sequences
        a_seq <- c(a_seq, A)
        b_seq <- c(b_seq, B)
        x_seq <- c(x_seq, x)
        f_seq <- c(f_seq, f)

        if (H(x)*H(B) < 0){#right push case
          A <- x
          B <- B
        }
        else if (H(x)*H(B) == 0){#rare root finding case
          print(paste("The root is ",x))
        }
        else{#left push case
          A <- A
          B <- x
        }
        #counter
        i <- i+1
      }
      return(list(
        A = a_seq, B = b_seq, X = x_seq, f = f_seq, Estimate = x
      ))
    }
    else {print("Bisection Algorithm Fail: Try a different interval!")}
  }
```

```
  else{print("Error: Improper Interval")}
##end of function
}
```

Newton's Method Algorithm

```
## Newton's Method ##

## Must give input as expression in "x"
Newt <- function(expr, x0, tol = 1e-8, max_iter = 100){
  f_expr <- expr
  df_expr <- D(f_expr, "x")

  f <- function(x){
    eval(f_expr)
  }
  df <- function(x){
    eval(df_expr)
  }

  i <- 0
  x_seq <- c(x0)
  x_old <- x0

  for (i in 1:max_iter) {
    x_new <- x_old - f(x_old) / df(x_old)
    x_seq <- c(x_seq, x_new)

    if (abs(x_new - x_old) < tol) {
      break
    }

    x_old <- x_new
  }
  return(x_seq)
}

## Expression for Problem 2
f <- expression(x^2+x-1)
## Calling NM function
Newt(f, 1, 1e-6)
```